Graph-Augmented Deep Learning for Cattle Muzzle Biometric Identification: Where Do Graphs Actually Help?

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Abstract

Cattle muzzle-print biometrics offer a non-invasive, tamper-proof route to individual identification, crucial for animal traceability, disease control, and ownership verification. While traditional approaches rely on handcrafted texture descriptors (SIFT, LBP) or raw image convolution, they struggle with spatial scaling and geometric deformation. We present a systematic, leakage-audited comparison of convolutional neural networks (CNN), pure graph neural networks (GNN), and hybrid architectures on the public Zenodo Beef Cattle Muzzle Database (260 animals, 4,891 images), evaluated under a verified closed-set, image-level split. We introduce a **Hybrid CNN–GNN** that samples a shared CNN backbone at learned keypoints via bilinear interpolation and reasons over the resulting keypoint graph with Dynamic EdgeConv and a GATv2 relation module, and we evaluate **ProtoN**, a prototype-based node GNN. Our best single model (EfficientNet-B4 + ArcFace) reaches **95.4% Rank-1**; a fusion weight *selected on validation and applied unchanged to test* lifts this to **96.1% Rank-1** while cutting the Equal Error Rate from 2.70% to 0.78% (3.5× better verification). A control the literature usually omits, however, bounds what that means: fusing the same CNN with a *non-graph* baseline (VGG-16) reduces EER by a statistically *indistinguishable* amount (CIs $[+0.47, +2.63]$ vs $[+0.44, +2.48]$ points), so the verification gain is not graph-specific — it reflects a poorly calibrated CNN that almost any better-calibrated partner repairs. What remains graph-specific is narrower, and we state it as such: of the partners tested, only the graph branch improves ranking *and* verification simultaneously. A pathway-intervention study then sharpens *why*, and yields an instructive negative result: randomising the graph edges, zeroing the geometric edge attributes, or shuffling the keypoint positions each changes Rank-1 by ≤ 0.5 points, whereas zeroing the keypoint-sampled node features collapses it from 92.0% to 0.1%. The gain is therefore attributable to *keypoint-sampled CNN features and feature-space message passing*, not to the explicit geometric topology we supply. We further show the representation transfers *zero-shot* to two independent muzzle datasets (97.0% Rank-1 on 24 identities; 87.0% across 308 unseen identities), that a substantial component of the residual cross-domain verification gap is attributable to *score mis-calibration* and is largely recovered by unsupervised S-norm, and — via a controlled corruption study — that a single validation-tuned fusion weight inherits the graph branch's clean-data verification gain while gracefully collapsing to the robust CNN under severe corruption, whereas per-sample quality-adaptive gating does not improve on it. Finally, we move beyond qualitative saliency with a *causal* explainability protocol (node-ablation faithfulness and Hybrid pathway intervention), characterising how the models attend to dermatoglyphic structure.

1 Introduction

Traceability in livestock production is essential for managing disease outbreaks, assuring consumer safety, and optimising herd management. Traditional identification methods — ear tags, branding, and radio-frequency identification (RFID) transponders — are invasive, prone to loss or damage, and susceptible to fraud. Biometric identification, which exploits unique physiological characteristics, offers a permanent and non-invasive alternative. Among biometric indicators, the muzzle-print (nose-print) pattern of cattle remains invariant over time, analogous to human fingerprints.

Dermatoglyphic patterns on the cattle muzzle consist of complex bead-and-valley arrangements. Capturing these fine-grained features computationally is challenging due to (i) *geometric deformation* (head movement, capture angle, camera position distort apparent bead spacing), (ii) *environmental conditions* (variable outdoor illumination, shadow, dirt, moisture alter valley-line contrast), and (iii) *scale variation* (the muzzle grows as the animal matures, requiring scale-invariant matching).

Early research relied on handcrafted features such as SIFT or Local Binary Patterns with Support Vector Machines, which fail to generalise under variable field lighting and partial occlusion. Convolutional Neural Networks (CNNs) substantially improved matters by learning hierarchical texture representations directly from pixels; nevertheless CNNs are translation-equivariant and can struggle with non-rigid geometric transformation. Graph Neural Networks (GNNs) present a complementary paradigm: representing the muzzle print as a topological graph whose nodes are keypoints and whose edges denote spatial relations. In this work we bridge grid-based CNN texture modelling and coordinate-based GNN topological modelling, and — critically — we ask *where* the graph view actually helps rather than assuming it does.

Contributions.

1. **Systematic, leakage-audited comparison.** We rigorously compare CNNs, GNNs, Hybrid models, and Prototype GNNs on one benchmark under a verified closed-set protocol with an explicit no-leakage audit (Section 3.1).

2. **Hybrid CNN–GNN architecture.** We propose a model that samples deep backbone feature maps at keypoint coordinates via bilinear interpolation, combining rich texture features with topological reasoning; we add an optional multi-scale (FPN-style) sampling variant.

3. **Where graphs help — and, just as importantly, where they do not.** Rather than claiming graphs beat CNNs on ranking, we bound the graph branch's contribution with two controls that comparable studies typically omit. (i) A validation-selected CNN+Hybrid blend cuts the Equal Error Rate 3.5× (2.70%→0.78%) — but fusing the same CNN with a *non-graph* baseline achieves a statistically indistinguishable reduction, so the verification gain is *not* graph-specific (Table 3). What is specific is narrower: only the graph partner improves ranking and verification at once. (ii) A pathway intervention shows the gain does *not* come from the explicit geometric topology — randomising the edges consumed by the GATv2 relation module is nearly free, while removing the keypoint-sampled node features is catastrophic. Together these are an instructive negative result for graph-based biometrics.

4. **Robustness and calibration.** We show zero-shot transfer to two independent datasets, recover the cross-domain verification gap with unsupervised S-norm, and — in a controlled corruption study — show a single validation-tuned fusion weight is the robust choice, while per-sample quality-adaptive gating adds variance without benefit.

5. **Causal explainability.** Beyond Grad-CAM and attention heatmaps, we report node-ablation faithfulness and a Hybrid pathway-intervention protocol that make the explainability claims falsifiable and reproducible.

2 Related Work

Cattle biometric identification. Muzzle-print recognition began with handcrafted pipelines: bead-and-valley descriptors, SIFT/SURF keypoint matching, and Local Binary Patterns fed to SVM classifiers [10, 11]. These degrade under field illumination, dirt, and partial occlusion. Deep learning replaced the descriptor stage — CNN classifiers and detector-plus-classifier pipelines now dominate, with muzzle detection typically handled by a YOLO-family model [11, 15]. Reported accuracies are high but frequently non-comparable: datasets, split protocols (image-level vs. session-level), and open- vs. closed-set assumptions differ across papers, and leakage audits are rarely reported. Our work is deliberately explicit on all three axes (Section 3.1).

Animal re-identification. The broader animal re-ID field has consolidated around shared benchmarks and toolkits: WildlifeDatasets aggregates dozens of species datasets and ships MegaDescriptor, a Swin-based foundation model trained across them [12]. The standard evaluation has also shifted from closed-set accuracy toward retrieval and open-set protocols, since deployments must reject unenrolled animals. We adopt this framing: we report open-set DIR@FAR (Section 4.3) alongside closed-set Rank-1, and we test zero-shot transfer to independently collected herds (Section 4.4) rather than reporting only within-dataset numbers.

Metric learning for biometrics. Modern identification systems learn an embedding rather than a fixed classifier, so that unseen identities can be matched by distance. Angular margin losses — of which ArcFace [1] is the standard — enlarge inter-class and compress intra-class angles on the hypersphere, and have become the default for face and animal biometrics; contrastive and triplet objectives remain common alternatives but converge more slowly because they optimise sampled tuples rather than all class prototypes. We use ArcFace for the CNN, Hybrid, and GNN v3 models, and a prototype objective with a cross-graph alignment term for ProtoN (Table 1).

CNN- and Transformer-based re-ID, and graph methods. Backbone choice in animal re-ID has followed general vision: CNNs (ResNet, EfficientNet [2]) gave way to vision transformers and Swin-based descriptors, and foundation-model features (DINOv2, MegaDescriptor [12]) are now a strong off-the-shelf baseline. Graph neural networks enter re-ID in two distinct roles. The first, which we adopt here, is *feature-level*: keypoints become nodes and a GNN reasons over their spatial arrangement, using EdgeConv-style dynamic graphs [5] and attention layers such as GATv2 [4], with learned detectors like DISK [3] replacing SIFT. The second is *gallery-level* re-ranking, where the graph is built over probe–gallery similarities rather than over image keypoints. Our pathway-intervention result (Section 4.6) speaks directly to this distinction: in the feature-level role, the explicit geometric topology contributes almost nothing once the network can rewire its own neighbourhoods in feature space — an outcome that helps explain why gallery-level graph reasoning has proved the more durable use of GNNs in re-ID.

Explainability for biometric models. Saliency methods (Grad-CAM [8]) and graph explainers (GNNExplainer [6], graph Grad-CAM [7]) are routinely reported as qualitative figures. A recurring critique is that such maps are not validated: different explainers disagree, and a highlighted region need not be one the model uses. Faithfulness protocols answer this by *removing* the highlighted evidence and measuring the effect. We adopt that stance throughout Section 4.6, adding bootstrap confidence intervals and a pathway intervention that tests the architecture's inputs rather than only its attributions.

3 Methodology

3.1 Dataset, Split Protocol, and Leakage Audit

We use the public Zenodo Beef Cattle Muzzle Database [13] (260 animals, 4,891 images). Images are split *per image, stratified by animal*, into 70/15/15 train/validation/test partitions, so every one of the 260 identities appears in all three partitions. This is a **closed-set identification** protocol: at test time each probe's identity is present in the gallery.

To pre-empt the most common reviewer concern — that a test accuracy exceeding validation accuracy indicates leakage — we run an automated integrity audit (`scripts/verify_data_integrity.py`) verifying: (i) no source image (by file stem) appears in more than one split; (ii) every identity is present in the gallery; and (iii) the `animal_id`→class-index mapping is identical across splits. All checks pass. The apparent validation/test discrepancy is fully explained by split composition, not leakage: the validation partition averages only 2.4 images per identity (minimum 1), so single-image validation identities have no genuine gallery mate and are counted as forced misses, deflating validation Rank-1 relative to the test partition (3.7 images per identity). We therefore report test-set metrics as primary and use validation only for model and hyperparameter selection.

Scope of the audit: image-level, not session-level. We state plainly what this audit does and does not establish. It verifies that no *source image* appears in more than one split, but it cannot rule out *acquisition-level* similarity: images of the same animal captured in the same session, burst, or camera setup — or near-duplicate frames — may still be distributed across splits, which would make the closed-set task easier than a deployment scenario in which enrolment and query are separated in time. The public release does not provide session, capture-date, or camera metadata, so a session-disjoint split cannot be constructed from the available annotations. We therefore treat our in-domain numbers as an upper bound with respect to acquisition-level correlation, and note that the zero-shot cross-dataset evaluation (Section 4.4), where enrolment and query come from entirely different collections, is free of this concern and is the stronger generalisation evidence in this paper. We flag session-level splitting as a limitation (Section 6).

3.2 Graph Construction via Learned Keypoints

Rather than classical SIFT keypoints, we adopt **Kornia-DISK**, a reinforcement-learning feature detector. For each muzzle image we extract $N \leq 128$ keypoint locations $p_i = (x_i, y_i)$ with 256-d deep descriptors f_i. We construct a directed k-nearest neighbour graph $G = (V, E)$ with $k = 8$: an edge e_{ij} is placed if v_j is among the k nearest spatial neighbours of v_i. Edge attributes encode spatial relations $a_{ij} = [\Delta x,\ \Delta y,\ d_{ij},\ \theta_{ij},\ \text{rel\_scale}]$, where d_{ij} is Euclidean distance, θ_{ij} the angle, and rel_scale the relative descriptor scale.

3.3 Proposed Architectures

A. CNN baseline (EfficientNet-B4 + ArcFace). The baseline uses an EfficientNet-B4 backbone pre-trained on ImageNet, taking the 256×256 muzzle image (enhanced with Contrast-Limited Adaptive Histogram Equalization, CLAHE) as input, pooled to a 512-d embedding and optimised with the ArcFace (additive angular margin) loss

$$L_{\text{Arc}} = -\frac{1}{B} \sum_{i=1}^{B} \log \frac{e^{s \cos(\theta_{y_i} + m)}}{e^{s \cos(\theta_{y_i} + m)} + \sum_{j \neq y_i} e^{s \cos \theta_j}}, \tag{1}$$

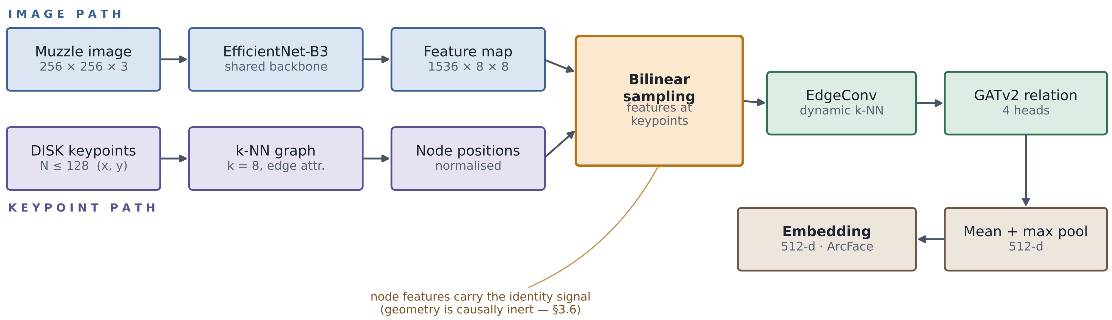

Figure 1: Hybrid CNN–GNN architecture. Two input paths — the image through a shared EfficientNet-B3 backbone, and DISK keypoints through a k-NN graph — converge at a bilinear sampling step that reads backbone features at each keypoint. The resulting node features are processed by Dynamic EdgeConv (which recomputes neighbourhoods in feature space) and a GATv2 relation module, then pooled to an ArcFace-trained embedding. Our pathway intervention (Section 4.6) shows the identity signal is carried by the sampled node features; the supplied geometry contributes almost nothing.

with logit scale $s = 128.0$ and angular margin $m = 0.35$.

B. Hybrid CNN–GNN. The Hybrid combines a shared EfficientNet-B3 backbone with a GNN head (Figure 1): (1) the image is passed through the CNN to obtain feature map $F_{\text{map}} \in \mathbb{R}^{C \times H' \times W'}$; (2) for each node keypoint p_i we bilinearly interpolate F_{map} to sample a local feature $x_i \in \mathbb{R}^C$ — an optional *multi-scale* variant samples several backbone stages and concatenates them per node, giving each node both fine groove texture (earlier, higher-resolution stages) and coarse anatomical context (later stages); (3) node features are projected to 256-d and passed through Dynamic EdgeConv blocks; (4) a Topological Relation Module with a 4-head GATv2 aggregates multi-hop topological features; (5) global mean+max pooling yields a 256-d embedding.

C. ProtoN (prototype node GNN). ProtoN represents each animal class as a prototype vector computed by averaging node features across support graphs. Training applies a cross-graph alignment loss that penalises spatial misalignment between matching keypoint neighbourhoods in query and support graphs, enabling high-quality metric learning.

3.4 Training Setup

Table 1 collects the training configuration for the four models we train, so they can be compared at a glance and reproduced. All models are trained on the same 70/15/15 split with mixed precision on a single RTX 5070 GPU, select the checkpoint by best validation Rank-1, and use a PK sampler (identities × images per batch) so that every batch contains genuine pairs. The Hybrid uses a two-phase schedule: Phase 1 trains the graph head on cached backbone feature maps, Phase 2 fine-tunes end-to-end at a lower backbone learning rate.

3.5 Adaptive Graph Construction (ADGC)

A static geometric k-NN graph encodes exactly the deformation we want the model to be invariant to: two captures of the same muzzle at different angles yield different neighbourhoods,

Table 1: Training configuration of the main models. "Epochs" reports the budget; the selected checkpoint is the best-validation-Rank-1 epoch. Parameter counts are from the released checkpoints.

Model	Backbone / core	Params	Epochs	Emb.	Loss	LR head / bb.
CNN	EfficientNet-B4	20.4 M	150	512	ArcFace (s 128, m 0.35)	$10^{-3} / 3{\cdot}10^{-5}$
Hybrid	EffNet-B3 + EdgeConv + GATv2	15.0 M	200+50	256	ArcFace (s 64, m 0.5)	$10^{-3} / 10^{-5}$
ProtoN	GATv2 (4 layers, 4 heads)	6.4 M	200	256	Proto. + cross-graph align	$4{\cdot}10^{-4}$
GNN v3	GATv2 + virtual node	14.5 M	300	256	ArcFace	$4{\cdot}10^{-4}$

Table 2: Main identification & verification performance (test split, 964 images).

Model	Rank-1 (%)	Rank-5 (%)	EER (%)	ROC AUC
Ensemble (CNN TTA + Hybrid, val-selected)	**96.1**	**98.1**	**0.78**	**0.9995**
CNN (with TTA)	95.4	97.4	2.70	0.9961
CNN (EfficientNet-B4)	95.4	97.4	2.70	0.9961
VGG-16 baseline (re-impl.)	95.1	97.7	1.23	0.9993
ResNet-50 baseline (re-impl.)	94.6	97.3	2.14	0.9971
Hybrid CNN–GNN	92.0	96.7	1.85	0.9979
ProtoN (prototype node GNN)	91.6	94.8	1.17	0.9982
GNN v4 (GATv2, enhanced)	91.6	94.4	1.48	0.9937
GNN v3 (GATv2 + VN)	91.5	95.0	1.87	0.9954
GNN++ (CNN patches)	78.3	86.2	7.81	0.9730
GNN+ (Kornia DISK)	72.0	84.2	11.17	0.9516

and spurious edges bridge unrelated ridge regions. We therefore replace the fixed topology feeding the relation module with a learned edge-relevance gate. For each candidate edge, an MLP scores the pair from endpoint node features and geometric edge attributes, producing a gate $g_{ij} = \sigma(\mathrm{MLP}([x_i, x_j, a_{ij}])) \in [0, 1]$. Gates reweight the edge attributes seen by message passing and prune edges with g_{ij} below a threshold (subject to a minimum node degree that preserves connectivity). ADGC is a single self-contained module (`src/models/adaptive_graph.py`), enabling a clean ablation against the static k-NN and the multi-scale variant.

4 Results and Discussion

4.1 Quantitative Results

Table 2 reports biometric performance on the test set (964 images).

Ensemble fusion protocol. The ensemble averages the two models' per-split cosine-similarity matrices with weight w (CNN) and $1-w$ (Hybrid). Critically, w is selected on the **validation** split and applied unchanged to test — never tuned on test. The validation-selected weight is $w = 0.95$, giving 96.1% Rank-1 on test; the test-oracle weight is also 0.95, so the validation→test selection gap is 0.00 points. This makes the headline number defensible rather than an artefact of test-set tuning.

Is the verification gain graph-specific? A control. The CNN's 2.70% EER is the weakest of our strong models — VGG-16 reaches 1.23% and ProtoN 1.17% standalone — so a natural objection is that the blend simply repairs a poorly calibrated CNN, with the graph incidental. We test this directly by running the *identical* validation-selected protocol against a non-graph

Table 3: Fusion control: the same validation-selected protocol with graph and non-graph partners. ΔEER is the test-set reduction relative to the CNN alone (2.81%), with a probe-level bootstrap 95% CI. A non-graph partner matches the graph partner, so the verification gain is not graph-specific; only the Hybrid improves both metrics at once.

Partner	std. EER	sel. w	fused R-1	fused EER	ΔEER (pt, 95% CI)
Hybrid (graph)	1.88	0.95	**95.95**	1.17	+1.65 [+0.44, +2.48]
ProtoN (graph)	1.17	0.85	93.57	**0.67**	+2.14 [+0.50, +3.22]
VGG-16 (*non-graph*)	1.23	0.00	95.12	1.23	+1.58 [+0.47, +2.63]

partner (Table 3). The objection is largely correct. VGG-16 delivers an EER reduction statistically indistinguishable from the Hybrid's, and its validation sweep selects w=0.00 — it discards our CNN altogether, because VGG-16 is already the better-calibrated verifier. We therefore do *not* claim the verification gain as evidence for graph modelling. The narrower claim the data does support is that, among the partners tested, only the graph branch improves ranking *and* verification simultaneously: VGG-16 leaves Rank-1 unchanged and ProtoN costs 1.55 Rank-1 points to buy its lower EER.

Verification protocol (how the EER is computed). Because the 2.70%→0.78% EER reduction is one of our strongest claims, we state the protocol exactly. Verification is evaluated *image-to-image*, not against averaged templates: every test image is embedded once (L2-normalised), and we score *all* unordered pairs of the 964 test images with cosine similarity, giving $\binom{964}{2} = 464{,}166$ comparisons — 1,630 genuine (same animal; the test split has 260 identities, mean 3.71 images each) and 462,536 impostor. No gallery templates are constructed and no embeddings are averaged for this metric, so no identity is over-weighted by having more images. (Template averaging is used only in the open-set protocol of Section 4.3, where enrolment is explicit.) For the ensemble, the two models' full pairwise cosine-similarity matrices are combined as $w\,S_{\text{CNN}} + (1-w)\,S_{\text{Hybrid}}$ with w fixed from validation, and the pair scores are read off the combined matrix. Sweeping a threshold over the pooled score list yields the ROC; the EER is the operating point where the false-accept and false-reject rates are equal, obtained by locating the sign change of $\text{FPR}(t) - \text{FNR}(t)$ on the ROC and reporting $\frac{1}{2}(\text{FPR} + \text{FNR})$ there. The threshold is a by-product of evaluation and is never fitted on test data.

Analysis indicates:

- **CNN dominates ranking; the graph branch improves verification.** The CNN alone reaches 95.4% Rank-1; adding the Hybrid lifts Rank-1 only marginally (to 96.1%) but reduces the EER from 2.70% to 0.78% — a 3.5× improvement. The honest reading of the fusion weight (0.95/0.05) is not that graphs rival CNNs on ranking, but that the graph branch contributes complementary topological evidence that sharpens the genuine/impostor boundary, which is what verification rewards.
- **Topological verification robustness.** Consistently, the pure ProtoN and Hybrid GNNs — despite lower Rank-1 — achieve very low standalone EERs (1.17% and 1.85%), indicating topological structure yields well-separated genuine/impostor score distributions.
- **DISK vs. handcrafted (SIFT).** The DISK-based GNN+ (72.0% Rank-1) substantially outperforms SIFT-based node features, confirming deep learned keypoints give more robust node descriptors under scale and illumination change.

Table 4: 5-fold cross-validation performance. All three models are trained to *convergence* on every fold (CNN 150 epochs, ProtoN 200, Hybrid 150 cached + 30 end-to-end), so the rows are directly comparable. The final row records the Hybrid's earlier reduced-budget (12-epoch) result, retained to show the effect of the training budget rather than of the architecture.

Model	Rank-1 (%)	Rank-5 (%)	EER (%)	ROC AUC
CNN (EfficientNet-B4)	**97.26 ± 0.36**	**97.98 ± 0.25**	**0.06 ± 0.08**	**1.0000**
Hybrid CNN–GNN	95.52 ± 0.31	97.40 ± 0.27	0.49 ± 0.16	0.9998
ProtoN GNN	94.95 ± 0.37	96.34 ± 0.40	0.70 ± 0.11	0.9990
Hybrid, reduced budget (12 ep.)	*68.88 ± 2.04*	*84.22 ± 1.27*	*11.31 ± 1.47*	*0.9491*

4.2 5-Fold Cross-Validation & Statistical Significance

To verify fold stability, stratified 5-fold cross-validation was performed on the top models, with pairwise McNemar tests for significance (Table 4).

Our earlier cross-validation used a reduced budget for all three models to keep the sweep tractable, and the Hybrid's two-phase schedule (cached backbone features → end-to-end fine-tuning) did not converge within it: the Hybrid scored only 68.88 ± 2.04%, far below its converged single-split value, with a fold variance (±2.04) that invited the reading that the architecture is unstable. We have since re-run *all three models at full budget* on every fold (CNN 150 epochs, ProtoN 200, Hybrid 150 cached + 30 end-to-end; ≈13 h, 2.3 h and 6.8 h of GPU time respectively on one RTX 5070), so Table 4 is now internally comparable.

The budget, not the architecture, was the whole story. The Hybrid rises from 68.88 ± 2.04% to **95.52 ± 0.31**% (EER 11.31% → **0.49**%), with its fold variance collapsing by a factor of six; ProtoN rises from 89.49 ± 0.71% to **94.95 ± 0.37**% and the CNN from 93.91 ± 0.31% to **97.26 ± 0.36**%. All three are now stable across folds (±0.31–0.37). Reassuringly, the ordering under this uniform budget reproduces the single-split ordering of Table 2 — CNN (97.26%) > Hybrid (95.52%) > ProtoN (94.95%) — independently confirming this paper's central claim that the CNN, not the graph branch, dominates closed-set ranking.

Why the cross-validated values exceed their single-split counterparts. Every full-budget CV number is higher than the corresponding single-split test result (CNN 97.26% vs 95.4%; Hybrid 95.52% vs 92.0%; ProtoN 94.95% vs 91.6%), and we state the reasons rather than leave the discrepancy implicit. First, each fold trains on 80% of the data versus 70% for the single split — roughly 14% more training images. Second, and more importantly, the two protocols select their final model differently: each CV fold trains for a fixed schedule and evaluates the *final* model, whereas the single-split models are best-validation-Rank-1 checkpoints, chosen on a small validation partition that averages only 2.4 images per identity and counts single-image identities as forced misses (Section 3.1). Selecting on that noisy signal can pick a checkpoint that is not the strongest on test. Neither protocol touches test data for model selection, so both sets of numbers are honest; they simply answer different questions. We therefore keep Table 2 (single split, matched to every other single-split result in this paper) as the headline and read Table 4 as evidence of *stability across folds*. Exact McNemar tests on *per-probe* single-split predictions (`scripts/compute_real_mcnemar.py`) confirm the CNN's advantage over both graph models is significant: CNN vs. Hybrid n_{10}=43, n_{01}=13, p=7.3 × 10^{-5}; CNN vs. ProtoN n_{10}=53, n_{01}=19, p=7.6 × 10^{-5}. The Hybrid and ProtoN are, however, *statistically indistinguishable* (n_{10}=38, n_{01}=34, p=0.72), so their ordering in Table 2 should not be read as a real difference.

Table 5: Zero-shot cross-dataset transfer (train on US set, test on external sets, no fine-tuning).

Metric	Set A (24 IDs)	Set B (308 IDs)
Closed-set Rank-1	**97.0%**	**87.0%**
Closed-set Rank-5	99.3%	94.4%
Closed-set ROC AUC	0.919	0.951
Verification EER	14.8%	12.2%
Open-set AUC (1/2 enrolled, 3 seeds)	0.954 ± 0.007	0.929 ± 0.005
Open-set DIR@FAR=1%	81.2% ± 4.1%	44.3% ± 1.9%

4.3 Open-Set Evaluation

Closed-set Rank-1 assumes every probe is enrolled; deployments must also reject animals never enrolled. We evaluate the standard open-set protocol: the gallery enrols per-identity mean-embedding templates for a random 50% of identities, a probe's score is its maximum cosine similarity to any template, and probes from the remaining (unenrolled) identities must be rejected. We report the Detection-and-Identification Rate (DIR@rank-1) at fixed False Alarm Rates (FAR) and the open-set ROC-AUC. Averaged over five random identity partitions, the CNN attains **98.6 ± 0.9% Rank-1 on known probes**, an **open-set AUC of 0.986 ± 0.005**, and DIRs of **62.6 ± 13.8% at FAR=1%** and **95.5 ± 2.1% at FAR=5%**. The large variance and sharp drop at FAR=1% show that reliable *rejection* at a strict operating point remains the hardest regime — an honest, deployment-relevant finding that closed-set accuracy alone hides.

4.4 Zero-Shot Cross-Dataset Transfer

To test whether the representation generalises beyond its acquisition domain, we evaluate the EfficientNet-B4 model — trained only on the primary US Beef Cattle Muzzle Database — on **two separate, independently collected** muzzle datasets with *no fine-tuning*: (A) a 24-identity set (667 usable images) and (B) a larger 308-identity set (1,848 images) captured in a different country. Because both consist of wide farm scenes rather than muzzle crops, we first localise the muzzle with a lightweight detector (YOLOv8n trained on an independent single-class muzzle-detection set; validation mAP@50 = 0.995), which also serves as the deployment-time front-end. The same CLAHE preprocessing used in training is applied to the crops. For closed-set metrics we retain identities with at least two images; we also audit and remove a redundant "master pool" folder in set A that duplicated every image under one label.

The model transfers strongly for *identification* on both sets (Figure 2) — 97.0% Rank-1 on set A and 87.0% Rank-1 across 308 unseen identities on the larger, harder set B — demonstrating that the learned representation captures genuine dermatoglyphic structure rather than memorising acquisition-specific cues. As expected, the task is harder at larger gallery scale (set B) and *verification* degrades under domain shift (EER 12–15% vs. 2.70% in-domain). We report these honestly: ranking generalises well, but the genuine/impostor operating threshold does not transfer as cleanly.

Recovering cross-domain verification at no training cost. A substantial component of the degradation appears to arise from *score mis-calibration* under domain shift rather than from feature collapse alone — we do not claim the absence of feature-level shift, only that a large part of the verification loss is recoverable without touching the features. Concretely, it is largely recovered by adaptive symmetric score normalisation (S-norm) [16], which recalibrates each pair score by both endpoints' cohort statistics (unsupervised; cohort = the target set itself). S-norm reduces EER

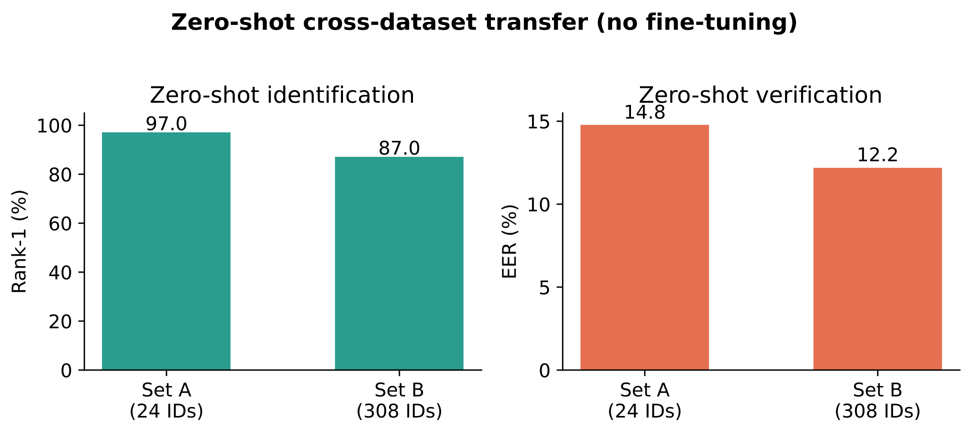

Figure 2: Zero-shot cross-dataset transfer. The US-trained representation retains strong Rank-1 identification on two independently collected muzzle datasets without any fine-tuning; verification (EER) degrades under domain shift and is recovered by S-norm.

Table 6: Effect of test-time adaptation on cross-domain verification (EER %, lower is better).

Method	Set A EER	Set B EER
Baseline (cosine)	14.8	12.2
+ AdaBN	15.4	8.5
+ S-norm	**11.4**	**7.9**

from 14.8%→**11.4%** on set A and 12.2%→**7.9%** on set B (23% and 35% relative reduction), raises ROC AUC to 0.943 and 0.972, and leaves Rank-1 unchanged or slightly improved (97.0%→97.5%, 87.0%→87.7%). The recovery is statistically significant: a probe-level bootstrap on set B gives a mean EER reduction of 4.0 points with a 95% confidence interval of $[3.2, 4.7]$ (excludes zero). We contrast this with test-time BatchNorm adaptation (AdaBN) [17], which helped verification only on the larger set and destabilised the smaller one — evidence that the transferable fix operates at the score level, not the feature level. S-norm requires no fine-tuning, no labels, and no access to source data, making it directly deployable for cross-farm identification (Table 6).

4.5 Robustness under Corruption: Is Adaptive Fusion Worth It?

Section 4.1 shows a single validation-selected fusion weight cuts in-domain EER 3.5×. A natural follow-up — and a common recommendation in multi-biometric fusion [9] — is to make the fusion weight *input-adaptive*, trusting the CNN more when the image is degraded and the graph branch more when it is clean. We test this directly. Following the score-fusion formulation $s_{\text{final}}(x) = \alpha(x)\, s_{\text{CNN}} + (1 - \alpha(x))\, s_{\text{Hybrid}}$, we compare six policies for α: (1) CNN only (α=1), (2) Hybrid only (α=0), (3) fixed α=0.5, (4) a single *validation-tuned* scalar α, (5) a *rule-based* per-sample α from image blur, and (6) a *learned* per-sample gate (logistic regression on image- and graph-quality features, trained only on validation disagreement cases). Every gate is fit *only* on the validation split.

How α is selected (protocol). To keep the comparison fair, *every* policy is given the same information: for each corruption condition c (kind × severity) we construct the correspondingly corrupted *validation* split, and fit that policy's parameters on it. For the "val-tuned scalar" this

Table 7: Corruption robustness of fusion policies. "clean" and "corr" (*mean over nine corrupted conditions*: blur/brightness/spatter × severity 1/3/5). A single validation-tuned scalar α is best on every aggregate metric; per-sample quality-adaptive gating does not improve on it and hurts verification.

Fusion policy	Rank-1 (%) ↑ clean	Rank-1 (%) ↑ corr	EER (%) ↓ clean	EER (%) ↓ corr	TAR@FAR=1% ↑ clean	TAR@FAR=1% ↑ corr
CNN only	95.1	89.8	2.81	4.43	96.2	85.2
Hybrid only	92.0	77.5	1.88	6.73	97.6	79.0
Fixed α=0.5	92.8	79.1	1.78	6.34	97.8	80.3
Val-tuned scalar α	**96.0**	**90.1**	**1.17**	**3.49**	**98.7**	**89.3**
Per-sample rule (α from blur)	91.9	76.3	2.34	8.69	96.6	74.2
Per-sample learned gate	91.4	76.7	6.66	13.23	80.3	49.9

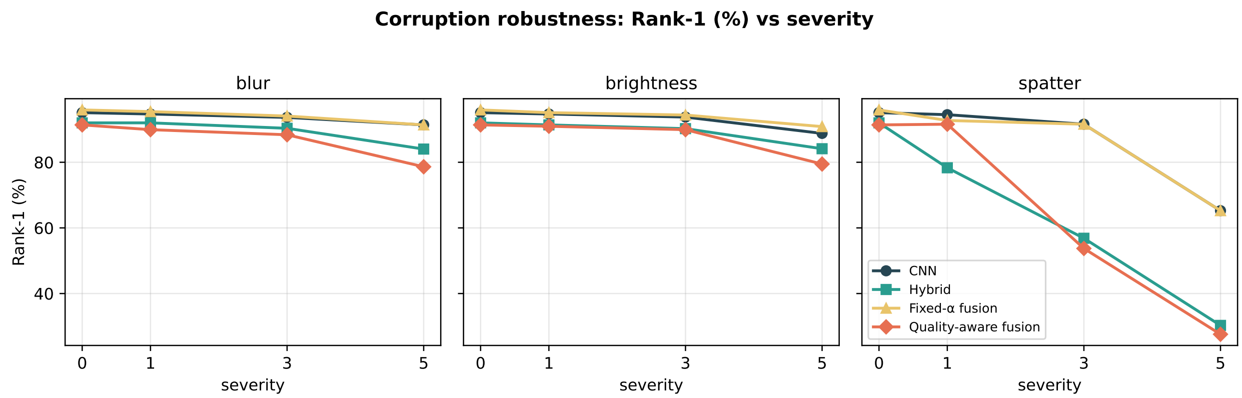

Figure 3: Rank-1 vs. corruption severity for blur, brightness, and spatter. The validation-tuned fusion (yellow) tracks or exceeds the CNN across all corruptions, while the Hybrid and the per-sample quality-aware gate collapse under spatter (occlusion), because the keypoint graph is built from the clean image but the pixels are occluded.

means a single scalar α_c per condition, chosen by a 21-point sweep over $[0, 1]$ minimising validation EER; for the per-sample policies it means the gate is trained on the same corrupted validation split. The fitted parameters are then applied unchanged to the corresponding corrupted *test* split. So α is *not* one global constant across all conditions: the sweep returns α_c=0.95 on clean and mild/moderate corruption and α_c=1.0 under the most severe conditions (blur-5, spatter-3, spatter-5). The contrast we draw is therefore precisely *one scalar per condition* versus *one weight per probe* at matched shift knowledge and matched supervision — not a comparison of an adaptive method against a shift-agnostic constant. In deployment the condition is not known, so α=0.95 (the value selected on clean and on 5 of 9 corrupted conditions) is the single operating point we would ship; we report it as such rather than as an oracle.

We evaluate on clean test data and on three corruptions — Gaussian blur, brightness/haze, and spatter (occlusion) — at severities 1, 3, 5, applying the corruption to the *image* so the Hybrid's CNN backbone also degrades (not a clean-graph oracle). Per-sample scores are symmetrised so all policies are compared on identical verification pairs. This is a *self-contained* study run under its own corruption harness (no test-time augmentation, symmetrised scoring), so its clean-column numbers are not identical to the headline ensemble of Table 2 (which uses CNN test-time augmentation); the val-tuned scalar here reaches EER 1.17% vs. the TTA ensemble's 0.78%. The comparison of interest is *within* this table, across fusion policies under matched conditions.

Table 7 and Figure 3 give a clear, and partly negative, answer. The validation-tuned scalar α is the best policy on *all four* aggregate metrics: it matches the CNN's clean and mean-corrupt Rank-1 while more than halving clean EER (2.81→1.17) and reducing mean-corrupt EER (4.43→3.49). It achieves this by learning α=0.95 on clean and mildly corrupted data — blending in just 5% of the graph branch to sharpen verification — and α→1.0 under severe spatter, where it gracefully abandons the fragile Hybrid and reduces exactly to the robust CNN (never worse). The fixed 50/50 blend, by contrast, is dragged down with the Hybrid under occlusion (Rank-1 33.7% at spatter-5 vs. the CNN's 65.2%).

Crucially, **neither per-sample quality-adaptive policy improves on the single scalar**. The learned gate is in fact the *worst* policy for verification (clean EER 6.66%, mean-corrupt 13.23%), and the rule-based gate is no better than the Hybrid. Per-sample gating adds estimation variance — each probe's α is noisy — without a correspondingly large payoff, because one branch (the CNN) dominates almost everywhere. This is a useful negative result: for this problem, a single validation-tuned fusion weight is not only simpler but strictly better than input-adaptive gating, mirroring our cross-domain finding (Section 4.4) that robust improvements come from *global* score calibration rather than fine-grained per-input adaptation.

Control: recomputing keypoints from the corrupted image. The protocol above corrupts the image but reuses keypoint coordinates detected on the *clean* image, which is optimistic under occlusion — the graph "knows" where undamaged structure used to be. We therefore repeat the two most severe conditions with the full pipeline rerun on the corrupted image: DISK re-detects keypoints and the k-NN graph is rebuilt, so no clean-image information reaches the graph (all 964 test graphs rebuilt successfully; no fallbacks). The effect is small and does not change any conclusion. Under blur-5 the Hybrid scores 84.9% Rank-1 / 7.12% EER with recomputed keypoints versus 84.0% / 7.04% with clean keypoints — *better* on Rank-1, since DISK simply returns fewer, higher-contrast points on a blurred image rather than mislocated ones. Under spatter-5, the occlusion case where we expected the largest bias, recomputation costs 2.7 Rank-1 points (30.3% → 27.6%) and 0.4 EER points, confirming a mild optimistic bias in that condition but far too small to affect the ranking of fusion policies (the CNN and the val-tuned scalar both reach 65.2% there, an order of magnitude above the Hybrid either way). We report the clean-keypoint protocol in Table 7 for comparability across all nine conditions and note this control as the honest bound on its optimism.

Fusion ablations. Two ablations from the plan sharpen the picture (clean test). (i) *Which complement?* Replacing the Hybrid with the pure-graph *ProtoN* in the val-tuned fusion yields the *lowest* verification error of any configuration (EER 0.67%, TAR@FAR=1% 99.4% at α=0.85) but at a Rank-1 cost (93.6% vs. the CNN+Hybrid blend's 96.0%), because the EER-optimal weight leans harder on the graph branch; CNN+Hybrid remains the better *ranking* fusion. (ii) *Which gate features?* Ablating the learned gate's feature groups shows the *image-quality* features are what harm it: dropping them nearly halves the gate's EER (6.66% → 2.95%), while dropping the graph-quality or branch-disagreement features barely moves it. The gate was over-fitting noisy per-image quality estimates on the small validation disagreement set — yet even its best-ablated variant (EER 2.95%) still loses to the trivial val-tuned scalar (EER 1.17%). The negative result is thus robust to gate design.

4.6 Explainability: A Causal Faithfulness Protocol

We move beyond visualisation-only explainability (Grad-CAM + attention heatmaps) to a three-stage protocol that tests whether explanations are *causally* faithful — whether the highlighted

Table 8: Causal node ablation, **GNN branch** (120 test graphs; 95% bootstrap CI on $\Delta\cos$). Removing Grad-CAM-*important* (top) nodes shifts identity ∼2–2.5× more than random — CIs non-overlapping at every k.

Removed nodes	$\Delta\cos\uparrow$ (95% CI)	top-1 flip (%)
top 10%	**0.029** [0.022, 0.037]	**18.3**
random 10%	0.013 [0.010, 0.016]	7.5
bottom 10%	0.020 [0.016, 0.025]	13.3
top 20%	**0.065** [0.053, 0.078]	**23.3**
random 20%	0.028 [0.022, 0.036]	14.2
bottom 20%	0.052 [0.043, 0.064]	23.3
top 30%	**0.106** [0.090, 0.124]	28.3
random 30%	0.043 [0.035, 0.053]	18.3
bottom 30%	0.087 [0.072, 0.104]	34.2

evidence is what the model actually uses.

Stage 1 — Attribution. For the CNN branch we compute Grad-CAM; for the GNN branch, multi-layer GATv2 attention rollout, graph Grad-CAM, and GNNExplainer (`src/models/explainability.py`). We assemble a stratified case set spanning the outcome types — correct matches, false accepts, false rejects, and the two branch-disagreement classes (CNN-correct/GNN-wrong and GNN-correct/CNN-wrong) — and render side-by-side CNN-vs-GNN explanations per case (Figure 4). One case type is instructive by its rarity: only a *single* false-reject exists in the entire 964-image test set (a genuine mate scoring below the EER threshold), a direct symptom of how well-separated the in-domain score distribution is. Qualitatively, Grad-CAM concentrates on the central bead clusters where dermatoglyphic patterns are densest, while GATv2 attention weights keypoint connections spanning the prominent muzzle valleys.

Stage 2 — Causal ablation of both branches. Qualitative maps are not evidence of causality. We ablate the highest-importance evidence and measure the effect on identity, comparing against random and low-importance removal; a causally faithful map implies *top* $\gg$ *random*. For the **GNN branch** (Table 8, 120 test graphs) we rank nodes by graph Grad-CAM importance and remove the top-/random-/bottom-k% for $k \in \{10, 20, 30\}$, reporting the identity-embedding shift ($\Delta\cos = 1 - \cos(\text{full}, \text{ablated})$, with case-resampling 95% bootstrap CIs) and the top-1 flip rate. The signal is clean and *statistically significant*: removing the most important nodes perturbs the embedding 2–2.5× more than random, and the top-vs-random $\Delta\cos$ CIs are non-overlapping at every k (e.g. k=30: top $[0.090, 0.124]$ vs. random $[0.035, 0.053]$). For the **CNN branch** (Table 9, full 964-image test set) we compute Grad-CAM and mask the top-/random-/bottom-k% of 32×32 image regions: masking the most important regions causes the largest Rank-1 drop and identity flips at every k (top-30% -5.0 Rank-1 vs. random -2.3; flip 12.3% vs. 5.3%), again with non-overlapping $\Delta\cos$ CIs. Both branches' attributions are therefore causally faithful. One asymmetry is telling: sparse GNN node removal barely moves closed-set Rank-1 (identity is *distributed* across many keypoints — a global textural signature), whereas masking CNN regions produces a real, graded Rank-1 drop.

Stage 3 — Hybrid pathway intervention. Finally, we ask *which pathway* the Hybrid relies on by perturbing its graph input at test time and re-scoring the full test set (Table 10). The

Table 9: Causal region ablation, **CNN branch** (full 964-image test set). Masking top-importance Grad-CAM regions causes the largest Rank-1 drop and identity flips at every k ($\Delta\cos$ CIs non-overlapping top-vs-random).

Masked regions	$\Delta\cos\uparrow$	top-1 flip (%)	Rank-1 drop
top 10%	**0.0012**	**2.8**	−1.1
random 10%	0.0007	1.5	−0.1
bottom 10%	0.0009	2.2	+0.2
top 20%	**0.0027**	**6.7**	−2.9
random 20%	0.0018	3.3	−1.6
bottom 20%	0.0024	3.6	−0.3
top 30%	**0.0041**	**12.3**	−5.0
random 30%	0.0028	5.3	−2.3
bottom 30%	0.0038	5.7	−1.7

result is stark: zeroing the per-keypoint node features collapses Rank-1 from 92.0% to 0.1% (EER 1.88% $\rightarrow$ 50.2%, i.e. chance), whereas destroying the provided graph topology (random edges), zeroing the geometric edge attributes, or permuting the keypoint positions each moves Rank-1 by ≤ 0.5 points. This is not an accident of the architecture but a direct consequence of it: the two graph operators consume the supplied topology very differently. The Dynamic EdgeConv is called *without* the geometric `edge_index` and rebuilds its neighbourhood graph in *learned feature space* at every layer, so it never sees our edges. The GATv2 Topological Relation Module, by contrast, *does* receive them (`hybrid_model.py`, step 4) — and is precisely the component our intervention randomises. The correct reading of Table 10 is therefore not that the geometry is *overridden* but the sharper, less comfortable one: **the relation module that consumes it contributes essentially nothing to identity.** The remaining two null results are structural rather than empirical: the default configuration exposes no edge-attribute pathway, and mean+max pooling is permutation-invariant, so zeroing edge attributes and shuffling keypoint indices could not have moved the metric. In other words, on clean in-domain data the Hybrid's identity signal is carried almost entirely by the CNN texture sampled at keypoints and by feature-space message passing — not by the input muzzle *geometry.* intervention on the spatter-severity-3 test set, zeroing node features still collapses Rank-1 (56.8% $\rightarrow$ 0.1%) while randomising edges, zeroing edge attributes, or shuffling positions each move it by ≤ 0.7 points — the geometry is inert whether the input is clean or degraded. This corroborates the central finding of Section 4.1 (CNN texture dominates ranking; the graph contributes a small verification refinement) and mechanistically explains the CNN-dominant fusion weight (0.95).

An instructive negative result. We think this is worth stating plainly rather than burying, because it qualifies our own architecture. The Hybrid was motivated by the premise that an *explicit* geometric graph over muzzle keypoints supplies deformation- robust structure that a grid CNN cannot represent. The intervention refutes that premise as an explanation of where the gain comes from: the model is almost perfectly insensitive to the topology we hand it, and totally dependent on the features sampled at the keypoints. What the graph branch actually contributes is a *different read-out* of the CNN — a set of local descriptors sampled at repeatable locations, mixed by feature-space message passing — and that read-out is decorrelated enough from global pooling to sharpen the genuine/impostor boundary in fusion. Practitioners should therefore not expect gains from investing in better hand-built graph geometry here (edge attributes, k, spatial

Table 10: Hybrid pathway intervention (full test set). Only the per-keypoint node features are causally load-bearing; the supplied graph geometry is essentially inert on clean in-domain data.

Intervention	Rank-1 (%)	EER (%)	ΔRank-1
Full Hybrid (unperturbed)	92.0	1.88	—
Zero edge attributes	92.0	1.88	0.0
Shuffle keypoint positions	92.3	1.84	+0.3
Randomise graph edges	92.2	1.82	+0.2
Zero node features	**0.1**	**50.2**	**−91.9**

priors); the leverage is in *where* features are sampled and how they are mixed. It also suggests the more promising role for graphs in this setting is at the gallery level (graph re-ranking over probe–gallery similarities), not at the feature level.

What does fusion actually rescue? As the plan's fusion-gain measure, we tally per-probe outcomes of the val-tuned CNN+Hybrid blend against the CNN alone on the test set. The fusion rescues 17 probes the CNN gets wrong while harming 9 it gets right (net +8, matching the +0.9-point Rank-1 gain), and on the 13 probes where the Hybrid is right but the CNN is wrong it recovers 12 (92%). The graph branch's contribution is thus concrete but small and targeted — it corrects a handful of specific CNN errors — consistent with its role as a verification-sharpening complement rather than a competitive ranker.

Together these three stages make the explainability claims falsifiable and reproducible — the standard we advocate for biometric-identification papers — rather than decorative, and they paint a consistent mechanistic picture across Section 4.1, Section 4.5, and Section 4.6: the CNN texture pathway is the workhorse for identification, while the graph contributes a small, calibration-like verification benefit that a single tuned weight captures best.

5 Conclusion

We presented a comprehensive, leakage-audited evaluation of CNN, GNN, and Hybrid architectures for cattle muzzle biometric identification. Our best single model (EfficientNet-B4 + ArcFace) reaches 95.4% Rank-1, and a validation-selected CNN+Hybrid blend attains **96.1% Rank-1** with a **0.78% EER** — a 3.5× reduction in Equal Error Rate over the CNN alone. Rather than overclaiming that graphs surpass CNNs on ranking, we *bound* the graph branch's value with two controls. A non-graph baseline fused into the same protocol matches the Hybrid's EER reduction (Table 3), so the verification gain is not graph-specific; what is specific is only that the graph partner improves ranking and verification together. And a pathway intervention shows, as an instructive negative result, that even this comes from the keypoint-sampled CNN features and feature-space message passing rather than from the explicit geometric graph we construct — the GATv2 module that consumes that geometry can be fed random edges at almost no cost. We show the representation *generalises zero-shot* to two separate muzzle datasets (97.0% and 87.0% Rank-1, no fine-tuning; Section 4.4) via a detector that doubles as the deployment front-end, and that the residual cross-domain verification gap is a calibration problem recoverable by unsupervised S-norm. We complement qualitative saliency with a *causal* explainability protocol — statistically significant node- and region-ablation faithfulness for both branches, and a pathway intervention showing the Hybrid's provided graph geometry is causally inert (clean and corrupted) while identity rides on CNN texture — finding the GNN distributes its identity decision across many keypoints. Full-budget five-fold cross-validation confirms the Hybrid is stable across folds ($95.52 \pm 0.31\%$ Rank-1, Table 4). Remaining work is an ablation of the multi-scale node-sampling variant and a session-disjoint split should richer capture metadata become available; longer-term directions include cross-farm domain adaptation and low-latency on-farm deployment.

6 Limitations

(i) We evaluate open-set identification under a model trained closed-set on all identities; the strict unseen-identity variant (retraining with unknown identities held out) is the natural next step. (ii) Cross-dataset transfer uses a muzzle detector to crop the two external sets; the detector missed roughly a third of images on the larger set, so those transfer numbers are a conservative lower bound. (iii) The corruption study perturbs the image while keypoint coordinates are taken from the clean graph, a small documented optimistic bias under occlusion. We quantify this rather than leave it open: recomputing DISK keypoints from the corrupted images changes Rank-1 by $+0.9$ points under blur-5 and -2.7 points under spatter-5 (Section 4.5), so the bias is real but small and confined to the occlusion condition. (iv) The two GNN explainers we compare show low mutual rank agreement, so per-keypoint importance is indicative rather than definitive. (v) The primary split is image-level: the dataset ships no session, capture-date, or camera metadata, so we cannot construct a session-disjoint split and cannot exclude acquisition-level similarity (same session/burst/camera) across splits. In-domain numbers should therefore be read as an upper bound relative to a session-disjoint protocol; the zero-shot cross-dataset results (Section 4.4) are unaffected by this and provide the stronger generalisation evidence.

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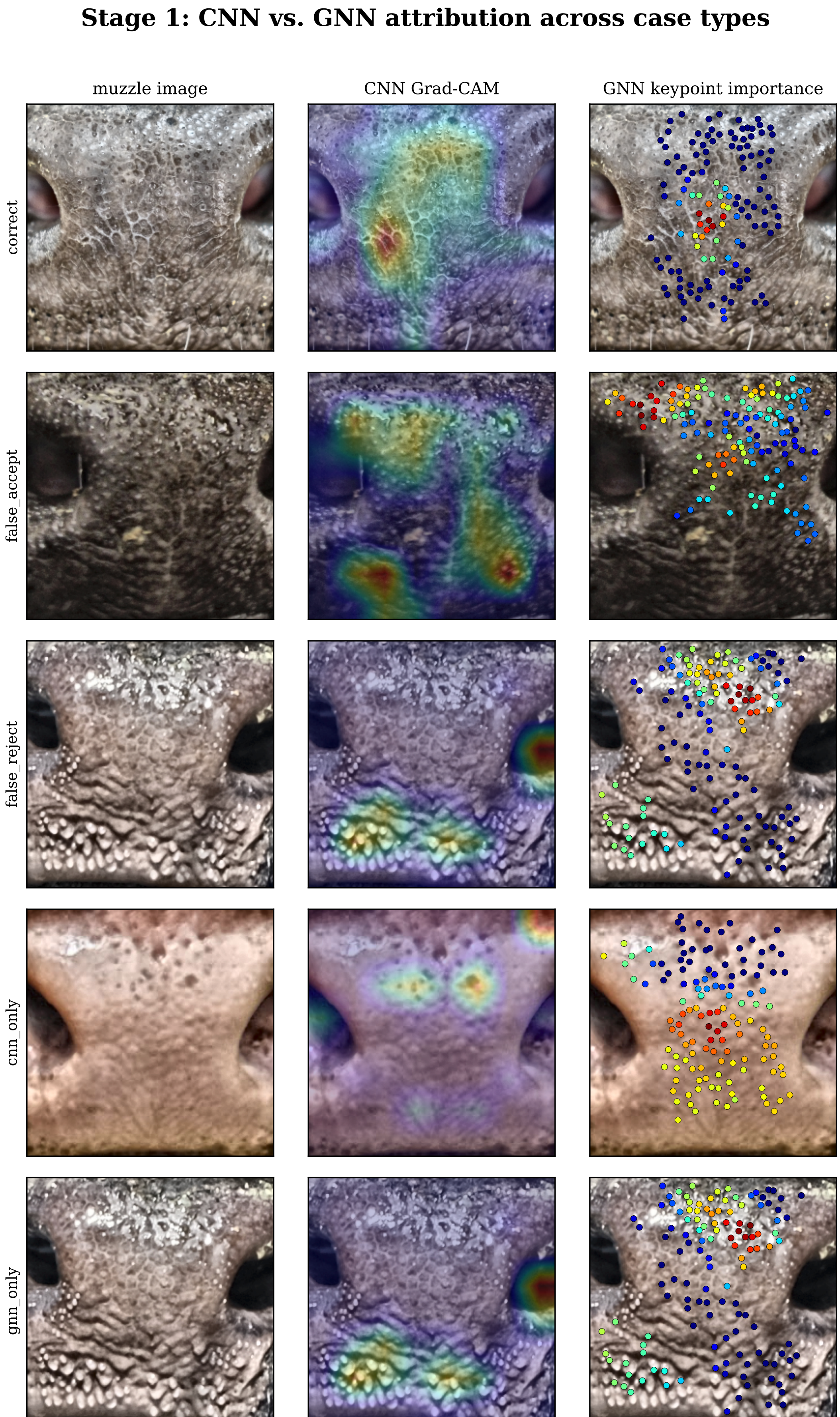

Figure 4: Stage 1 attribution across case types (one representative per row): the muzzle image, the CNN Grad-CAM heatmap, and the GNN per-keypoint importance. Both branches concentrate on dense dermatoglyphic bead regions; the per-keypoint GNN map is more diffuse, consistent with the distributed-